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Chatbots and Procrastination: Identifying and Mitigating Risks

A Thematic Analysis of Expert Interviews

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Abstract

Procrastination is widespread and closely tied to well-being, stress, and academic and work performance. This explorative study investigates potential risks of using large language model chatbots to treat procrastination, as well as design approaches that could mitigate these risks. Experts in three areas, psychology, language technology, and human-computer interaction, were interviewed, and thematic analysis was conducted on the resulting material. Five themes emerged, concerning individual adaptation, engagement and motivation, alignment and feedback, social factors, and anthropomorphism. Key risks include alignment conflicts between general and specialised chatbots, loss of social accountability, anthropomorphism-induced overtrust, and over-generalisation in treatment. Proposed design responses include gradual user modelling, predefined personality presets, simulated accountability mechanisms, language adaptation, and redirection to alternative treatments when appropriate. The study contributes to understanding the specific challenges of using chatbots for procrastination reduction and suggests that while some risks are inherent to the technology, others are addressable through design and warrant consideration by developers.

Keywords: Procrastination, chatbots, large language models, AI risks, thematic analysis, expert interviews, anthropomorphism, personalisation

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1. Introduction

The 30th of November 2022m OpenAI launched ChatGPT. In the following months the user counts skyrocketed. By January 2023 the service had an estimated 100 million users, which made it the fastest growing application in history up to that point (Hu, 2023). Chatbots have been adopted at a pace unprecedented in consumer technology, prompting widespread public and academic attention. Multiple chatbots from different companies have emerged, and they have become a given part of our society. When a technology field develops as rapidly as artificial intelligence has, it is crucial that we quickly start to understand the ways in which it will affect our society, and what risks may emerge. Concerns have been raised when it comes to for example security, copyright, and education (Farina & Lavazza, 2023). These chatbots can be used in a wide variety of ways, and some potential use cases include therapy and productivity facilitating advice.

One way in which to increase productivity is to reduce procrastination. Procrastination can be defined as the voluntary delay of an intended course of action despite expecting to be worse off for the delay (Steel, 2007). The phenomenon is widespread and causes significant problems for a substantial amount of people, especially amongst students (Rezaei-Gazki et al., 2024). This causes problems on a societal level both in the way of giving rise to mental health issues, but also because it delays productivity, which on a large scale could have cascading effects (Stead et al., 2010). There are, of course, many possible ways of addressing this problematic behaviour, for example psychiatric methods such as cognitive behaviour therapy.

These new chatbots could be used as an alternative method, for example in giving answers to questions and real-time advice. Users could provide information about their problems and get help in the form of text from these tools. Consider a person that asks a chatbot ‘What strategies can I employ to get started with my project?’, followed by a response by the bot that tries to address the question. Potential benefits include faster, cheaper and more available help for anyone who might need it. People with less serious problems might get help even though they don’t need regular therapy treatment. The help could be available instantly even outside of regular working hours. This might seem like an excellent opportunity, but it is important to also consider potential risks of this type of treatment. Generative AI systems are known to produce confidently stated misinformation, to reflect biases present in their training data, and to lack the nuanced emotional understanding required for sensitive contexts (Ji et al., 2023) (Hagendorff, 2024). Whether and how such risks manifest when chatbots are used to help users address procrastination has not been examined. That is what this paper sets out to do.

Aim

The purpose of this study is to gather early insight on how current general purpose chatbots can be made safer when it comes to using them in personal development, specifically through the lens of using them to reduce procrastination. The aim is to identify the risks associated with this kind of usage, as well as potential workarounds that developers and users can implement in specialised purpose-specific chatbots.

Research Questions

1. What are some potential risks associated with using generative artificial intelligence in the form of chatbots for reducing procrastination?
2. How could these potential risks be mitigated, both by users and developers?

Delimitations

This thesis is specifically focused on chatbots in the form of large language models, based on the transformer technology. The analysis will have general-purpose chatbots such as ChatGPT as its object of study but will extend to design implications for purpose-built tools (still generative LLM-chatbots). It will therefore not try to identify the risks associated with AI in general, generative or other. As previously mentioned, the main area of study is that of procrastination reduction, which means that other areas within productivity or personal development will not be thoroughly investigated. The study does not mean to generalize opinions of experts in general, nor establish all the associated risks as definitive truths, but rather aims to find potential risks.

Structure of the Thesis

The background section following aims to equip readers with the information they need for subsequent chapters. It delves into procrastination, chatbot technology and its interactions with humans and known risks associated with generative artificial intelligence. The methodology section both describes the used methods of expert interviews and thematic analysis and the motivations for why they were chosen. The data analysis section presents the mentioned thematic analysis of the interview material. Following that is a discussion of the thematic analysis in relation to the research questions, limitations of the study and potential avenues for future work. The conclusion means to summarize key findings and explicitly address the research questions.

2. Background

Introduction to the Background

This review aims to guide readers to a deeper understanding of the intersection between chatbot technology and psychological growth, which should increase understanding of subsequent chapters. First procrastination is explored through its definitions, causes, and treatment methods. This is followed by an investigation of the technology behind modern chatbots, and the intersection between chatbots and personal development. The next section highlights general risks associated with generative AI.

Understanding Procrastination

The definition that will be used in this thesis is the one that Steel (2007) put forth, where procrastination is "the voluntary delay of an intended course of action despite expecting to be worse off for the delay". It is of importance that the delay is voluntary, running late due to traffic is not procrastination for example. Likewise, it is not procrastination if the delay is not anticipated to have any drawbacks. It is important to note that delay is not in itself a synonym to procrastination. A central question then is why people procrastinate even though they expect to be worse off for doing so. Temporal Motivation Theory (TMT) offers a comprehensive explanation of procrastination by framing it through a mathematical equation that incorporates four key elements: **expectancy, value, impulsiveness,** and **delay** (Steel & König, 2006). Note that the formula is not meant for calculation and rather serves as an illustration of the relationship between the elements.

$$Motivation = \frac{Expectancy\ x\ Value}{1 + Impulsiveness\ x\ Delay}$$

According to TMT, the motivation to engage in a task is a function of these factors. The theory posits that an individual's motivation to complete a task increases when they expect they can successfully complete the task (high **expectancy**) and when the task is perceived as valuable or rewarding (high **value**). However, this motivation is tempered by the individual's level of **impulsiveness** and the **delay** before the task's rewards or consequences are realized. If a task is seen as unpleasant or difficult, an individual may doubt their ability to complete it successfully, leading to low expectancy. Similarly, if the rewards for completing the task are distant or seem insufficient (high delay or low value), the motivation to tackle the task soon diminishes. Impulsive individuals are more sensitive to reward delay, making procrastination more likely. Thus, TMT explains procrastination as a natural outcome when tasks are perceived to have uncertain outcomes, are less rewarding, or are scheduled far into the future, especially among those who are more impulsive.

Self-determination theory (SDT) offers another perspective on motivation, framing it as either being intrinsic or extrinsic (Deci & Ryan, 2000). Extrinsic motivation are those things that motivates us, stemming from other factors than ourselves, such as social pressure or physical rewards. Intrinsic motivation, on the other hand, is defined as that which comes from within our psyches and includes things such as genuine interest or curiosity. The theory argues that intrinsic motivation produces deeper, longer lasting and subsequently more effective motivation. Furthermore, three psychological needs are presented as crucial in cultivating motivation: autonomy, competence and relatedness. People are more motivated when they feel that they are in control of their decisions and actions, when they feel that they are capable in what they are doing, and when they experience social connection.

Readers might wonder why motivation-based theories are the focus of both frameworks, when procrastination is what is supposed to be modelled. The causes of procrastination are multifactorial. Short-term mood regulation for example, clearly contributes (Sirois & Pychyl,

2013). In TMT, which behaviour that is pursued is by definition a direct result of which behaviour is the most motivating. Within TMT's framework, other contributing factors are conceptualised as components of this calculation rather than as separate causes. This is why Steel (2007) can present TMT as a comprehensive framework for procrastination despite its motivational focus. SDT on the other hand is not explicit in defining motivation the same way. Neither is it explicitly a theory about procrastination. Its contribution is meant to inform design decisions about what kind of motivation an intervention should cultivate. Considering this especially it should be acknowledged that a complete account of procrastination would need to include other dimensions of the phenomenon. This is true even of TMT itself, which represents one framing among possible others. So, this thesis does not claim to give a full account on the phenomenon but rather provide enough to address the research questions.

Unfortunately, the research on how to treat procrastination is relatively lacking in quantity, but psychological treatment has been observed to provide a small benefit. Cognitive behavioural therapy (CBT) seems especially promising when it comes to a reduction in procrastination, but more research is needed to verify its effects (de Haas et al., 2025).

Chatbot Technology and Human Interaction

Deep learning, a cutting-edge subset of machine learning, uses multi-layered artificial neural networks to analyse various forms of data (Alzubaidi et al., 2021). These networks learn from large volumes of data by adjusting connections between what is in the field called nodes in response to the information processed. The technical details can become rather complex, and it is outside of the scope of this paper to give a detailed explanation of the inner workings of this technology. For readers unfamiliar with the area, it should suffice to know that this approach has been an important part of enabling machines to perform complex tasks such as image processing and speech recognition.

Natural language processing (NLP), the technology underlying chatbots, has benefitted enormously from deep learning. It enables chatbots to understand, interpret, and generate language in a way that is contextually and semantically meaningful. The real breakthrough in NLP came with the introduction of transformers (Vaswani et al., 2017). Unlike their predecessors, transformers use an "attention mechanism" that allows the system to weigh and focus on different parts of the input data independently of their sequence in the data. This ability to process data points in parallel, rather than sequentially, significantly enhances the effectiveness of the model in understanding language (Aslam, 2023). These models facilitate a deeper understanding of context, allowing chatbots to maintain more coherent and contextually relevant dialogues with users. As a result, modern chatbots powered by transformers are increasingly sophisticated, capable of providing customer support, personalised interaction, and even engaging in complex discussions on a wide range of topics.

One of the other main driving factors in enabling modern chatbots to be as useful as they are is training through human feedback. Christiano et al. (2017) describe how traditional training of reinforcement learning systems is done by evaluating weights against a reward function. As systems and their purposes grow more advanced it becomes harder to define a reward function mathematically. The solution is to instead do the evaluation against human feedback. The system produces an output which then a human rates, for example as good or bad. This can over time align the system with what humans actually want, even when that want is complex and hard to pin down. In terms of chatbots in particular, alignment is crucial in increasing truth and non-toxicity (Ouyang et al., 2022). The process of alignment is done through what is called fine-tuning, which means re-calibrating the weights of an already existing model after for example adding another output layer to the neural network.

In recent years, even before the transformer-based models, the integration of chatbots and conversational agents into the domains of mental health therapy and personal development has gotten significant attention. This is not an entirely new phenomenon, already in the 1960s

we had programs such as ELIZA (Weizenbaum, 1966). ELIZA was quite a simple program, especially in comparison to what we have today, but it still managed to garner significant attention because people felt that their conversations with it were meaningful even though they understood the program's simplicity. Later research suggest that human-computer interaction is inherently social, and that this is not because of faulty beliefs but rather due to how deeply social humans are, which then makes the social response automatic (Nass et al., 1994).

Related to this is the human tendency towards anthropomorphism (attributing human characteristics to non-human entities) which is further amplified when those entities communicate in convincingly human-like language (Epley et al., 2007). The phenomena can be seen as being driven by three key factors. Firstly, we apply human like models on non-human entities because that is our most advanced, accessible and developed framework for understanding agents. Secondly, we anthropomorphise because we desire effective interaction, and we think that attributing human qualities to something helps us understand it, even though it might not. Thirdly, humans seek social connection, and anthropometrism can help in partly fulfilling that need.

Today, research indicates that chatbots are effective in delivering psychotherapeutic interventions, particularly for managing symptoms of depression and anxiety. Studies such as those summarized in the article "Chatbot-Delivered Psychotherapy for Adults With Depressive and Anxiety Symptoms" demonstrate that these digital interventions can significantly improve depressive symptoms, leveraging frameworks like CBT and problem-solving therapy (Lim et al., 2022).

In a similar fashion these technologies could also play an important role in personal development, particularly in aiding self-regulation and enhancing mindfulness. These tools could for example provide psychoeducational content that helps users understand and manage their psychological processes more effectively (Narynov et al., 2021). An already existing attempt to do this is the educationally centred chatbot StudyBuddy. It means to help users by offering structured reminders and breaking down larger tasks into smaller, more manageable objectives. This is then meant to enhance productivity and learning outcomes (Tian et al., 2021).

There are already a couple of examples of purpose-built procrastination chatbots in the literature. GanttBot is a Telegram-based chatbot that is designed to reduce procrastination for students working on capstone projects. It is integrated with project management software and has theoretical grounding in CBT. Rather than generative it is rule-based in its design and uses pre-defined messages to interact, coupled with tutor interaction through notifications. Which kind of interactions are used depends on where in a three phase process the student is in. Results were measured in a small cohort comparison study where procrastination was measured using overdue days, where the treatment group averaged 10 days whilst the control group 19 days, a difference reaching statistical significance (Pereira & Díaz, 2021).
A more recent study investigated the use of a semigenerative design called Moa. This chatbot was an integration into a to-do list mobile app, is also grounded in CBT and focused more on the psychological dimensions of procrastination (instead of just time management). Moa is semigenerative in its design, using a Korean GPT-2 variant within a pre-framed scenario. A second process running in parallel to the conversation was a procrastination factor model which identified the cause behind the procrastination and tailored advice accordingly. During a pilot randomised controlled trial with 85 participants the treatment group (which used the to-do app with Moa) showed significant improvements in time management, stress scores and irrational procrastination (Lee et al., 2025). So, specialised designs can include both constrained reminder systems and more ambitious therapeutic agents. Initial results from both studies are promising, but the evidence remains small.

Risks with Generative AI

One concern about generative AI in general is potential biases, where AI may replicate existing societal stereotypes and inequalities through biased training data (Hagendorff, 2024). If the training data suggests racial biases in job application evaluation for example, then the model will simply inherit that unless it has other incentives not do so. If the recruitment process is handled by AI to a larger and larger extent, then new training data which represent our societal improvements will be produced to a smaller and smaller extent.

Whilst training through human feedback definitely can be beneficial, there are also problems that can arise from it, and one of those is sycophancy (Sharma et al., 2024). Humans tend to prefer output which feels good emotionally over that which is true. This can subsequently result in models being overly agreeable, which can be a problem in situations where factual accuracy and truth matter more than emotions.

Amongst the most prominent issues with chatbots is that of hallucination (Ji et al., 2023). When we say that a model hallucinates, we mean that it is producing nonsensical or false language, that is linguistically correct and presented with seeming confidence. This is because fundamentally a chatbot has no real sense of true or false. This can lead to serious consequences in critical situations. Imagine a chatbot telling someone, with seeming confidence, that a particular mushroom they're describing is safe to eat, when it is in fact very poisonous. Hallucination is a direct constraint on when it is safe to use these models, they cannot be trusted in situations where factual accuracy is of paramount importance.

When it comes to specifically chatbots, despite advancements in natural language processing and machine learning, they currently lack the nuanced emotional understanding that humans possess. This limitation can lead to inadequate responses to users who may be in severe distress or experiencing complex emotional issues (Pham et al., 2022). There are also ethical considerations, including the necessity to ensure that chatbots used in personal development purposes adhere to ethical standards that respect user autonomy and consent, ensuring users are aware of what chatbots can and cannot do. There is also the question of privacy and security of user data, especially considering the potentially sensitive information when it comes to personal development (Narynov et al., 2021). While some studies have explored implementations of specialised chatbot-based interventions for procrastination, no research was found that specifically examines the *risks* associated with this use of general chatbots and the design implications for specialised ones.

3. Methodology

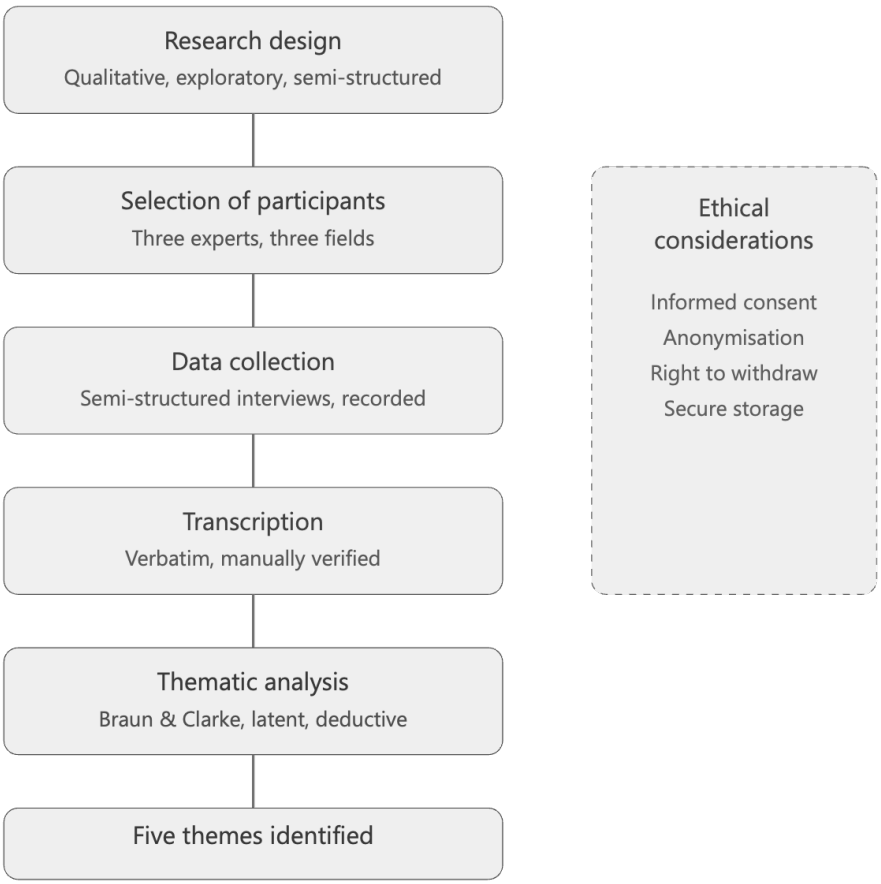


Figure 1: Methodology overview

Research Design

Because of the qualitative essence of the research questions, expert interviews were selected as the primary data collection method for this study. The goals of identifying and understanding, rather than quantifying, dictates a need for insights that are both authoritative and deeply informed, characteristics that expert interviews are uniquely positioned to provide. The nature of identifying risks requires more than speculative guesses; it demands insights with a foundation of expertise and experience. This necessity places limitations on the effectiveness of broader user surveys which, while valuable for capturing general sentiments, may lack the depth and specificity required to accurately identify potential risks. Through the conversational flow of an interview, it's possible to adapt questions in real-time, explore emerging ideas, and delve deeper into topics as they arise. This interactive dialogue is crucial for unearthing detailed insights and understanding the complex dynamics at play.

Selection of Participants

The process of selecting participants for this study was designed to get a comprehensive exploration of the research topic. The initial step involved an identification of the scientific disciplines most relevant to the study's focus areas. This was done through examining the literature, listing the most relevant areas, and then through a discussion with the study's first supervisor ranking them in terms of relevance. The field of psychology, relevant to the mechanics of procrastination, was seen as the most important. Following this, a detailed investigation was conducted to pinpoint an individual who possessed a profound expertise

within the area. Not before a participant was found and had agreed to participate did the search move on to the next discipline, which was that of language technology – the science behind chatbots. And when one expert was found there as well, the next chosen field was human computer interaction. When the third expert had been found, the next in line was an expert linguist. But no such expert was found before the study had to move on due to time constraints.

Each identified expert was contacted via email, which served two primary purposes. Firstly, the email provided an overview of the study, outlining its aims, significance, and the nature of their potential contribution. Secondly, the email extended an invitation for them to participate in the research, clearly stating that their involvement would be as a subject matter expert in an interview format. This approach to participant selection was guided by the principle of informed consent, ensuring that all experts were fully briefed about the study and its ethical considerations before agreeing to participate. The goal was to assemble a panel of participants whose expertise and insights would contribute significantly to the depth, quality, and relevance of the research findings.

Data Collection Procedures

The semi-structured interviews were planned in detail to elicit rich, informative insights into the use of chatbots in reducing procrastination, focusing particularly on identifying potential risks and their mitigation. In total, three interviews were conducted. Two of the interviews were conducted in person, while the third was conducted digitally, accommodating geographical and scheduling constraints. In preparation for these interviews, a comprehensive set of open-ended questions was formulated. See appendix B for the asked questions.

The first few questions meant to gather relevant background information that would contextualize the experts' insights. The rest of the questions meant to directly address the identification of potential risks associated with chatbot technologies in personal development and exploring viable workarounds. The questions were crafted to prompt the experts to think broadly and deeply about these issues, encouraging them to not only identify risks but also to speculate on possible solutions and preventative measures. In the end not all prepared questions were asked, because often the answers had been given in response to earlier questions. Other questions were rephrased during the interviews as to capture specific parts of the question that had not been answered earlier whilst accommodating for other parts of the questions that had. What is important to note is that the same questions were not asked to all of the experts. This choice was made because even though standardized questions would make the answers of the different experts more directly comparable to each other, it would make ascertaining relevant information of their particular area of expertise more difficult. Each question was developed with care to in conjunction with each other capture as many relevant aspects of the area as possible.

To enrich the discussions further, the questions were intentionally designed to facilitate open-ended responses. This approach was adopted to capture the breadth and depth of the experts' knowledge and perspectives, ensuring that valuable insights were not inadvertently overlooked. The flexibility of open-ended questions also allowed for the exploration of adjacent topics, providing a richer understanding of the context surrounding the identified risks. For each interview a final question of if there is anything that they feel is relevant to the topic at hand were asked, as to not miss anything crucial. Acknowledging the dynamic nature of conversation, the interviews were structured to be semi-open. While a prepared set of questions guided the discussion, the format was not rigid. The questions served more as topics of conversation than something that had to be answered directly. This flexibility ensured that if during an interview, new, relevant themes or ideas emerged, they could be pursued through follow-up questions. Essentially, the semi-structured format of the interviews combined with the strategic design of the questions ensured that the discussions were both comprehensive and focused.

Transcription Procedures

The interviews were transcribed manually. Following this the audio-files were deleted from the service to maintain the interviewee's privacy. The first draft was verbatim, preserving filler words, grammatically incorrect sentences, false starts and repetitions. Following this all information leading to easy identification of participants were deleted from the transcriptions. After this the transcription was carefully edited for readability, only making the smallest possible changes to achieve correct written text. This process included removing the artifacts of the verbatim transcriptions, such as filler words and false starts. Punctuation was placed where it seemed appropriate. These edits were made to improve readability for quotation purposes while preserving the meaning and substance of what each participant said.

Data Analysis

The method chosen for the interview data was thematic analysis, as it is described in the paper 'Using thematic analysis in psychology' (Braun & Clarke, 2006). The first step of the process entailed deep familiarisation with the data. Following this was a systematic coding of the whole dataset, each sentence and paragraph was analysed. Whenever an idea or expression was deemed as potentially relevant to the study's purpose it was highlighted and summarised as a code. The codes themselves were created during the process of coding, rather than predefined. As such, most codes were unique in their exact phrasing. All the codes were then written down on physical post-it notes and moved into different piles based on similarity and relevance to the research questions. Different combinations and organisations were tested through repeated comparison with the dataset. Some organizations were scrapped because the piles had too few notes each, which would have resulted in weak themes. Others because the resulting theme would have been too far detached from the research questions. Finally, five piles were in position, resulting in one theme each.

A contextualist point of view was taken on the data. In other words, the answers from the experts were treated as informed interpretations of a real-world phenomenon, rather than direct representations of that phenomenon itself. This distinction is important because experts from different fields may understand and describe the same phenomenon through different theoretical lenses, vocabularies, and assumptions. In this view, differences in descriptions are understood as partly reflecting these interpretative frameworks, while still referring to a shared underlying set of real-world conditions. This contrasts with a more strictly realist position, where such differences would more likely be interpreted as varying degrees of accuracy in representing an objective reality.

The analysis was primarily driven by the research questions rather than aiming to provide a comprehensive description of the entire dataset and can therefore be characterised as a deductive rather than inductive approach. In addition, the analysis aimed to go beyond explicit surface meanings in the data and identify underlying assumptions and patterns, corresponding to a latent level of analysis rather than a purely semantic one. In other words, the interpretation focused not only on what was explicitly stated, but also on the meanings and ideas underlying the experts' accounts.

Ethical Considerations

In addressing the ethical considerations of this study, focus was placed on the anonymity and privacy of the participants. To this end, all identifiable information has been omitted from the study's documentation, ensuring that details leading to the easy identification of participants remain confidential. Furthermore, no personal information gathered during the interviews has been stored in any digital format.

To safeguard the integrity of this ethical approach, all documents related to the interviews have been securely stored in a manner that severs any linkage to the personal identities of the

interviewed experts. The interviews, while recorded for accuracy and thoroughness, were subject to strict data handling protocols. Specifically, the audio recordings were used solely for the purpose of transcription. Upon the completion of this process, the audio files were permanently deleted to eliminate any potential risk of a privacy breach.

Additionally, prior to their participation, all interviewees were provided with detailed information about the study's objectives, the nature of their involvement, and how their data would be used and protected. Informed consent (see Appendix A for the digital version) was obtained from each participant, affirming their understanding and agreement to the terms of participation under the outlined ethical guidelines. This consent process also included assurances of their right to withdraw from the study at any point without any adverse consequences.

4. Results

The thematic analysis (see Section 3.4) generated five main themes. Transcripts can be found as appendixes C, D and E. P1, P2 and P3 refers to the three experts.

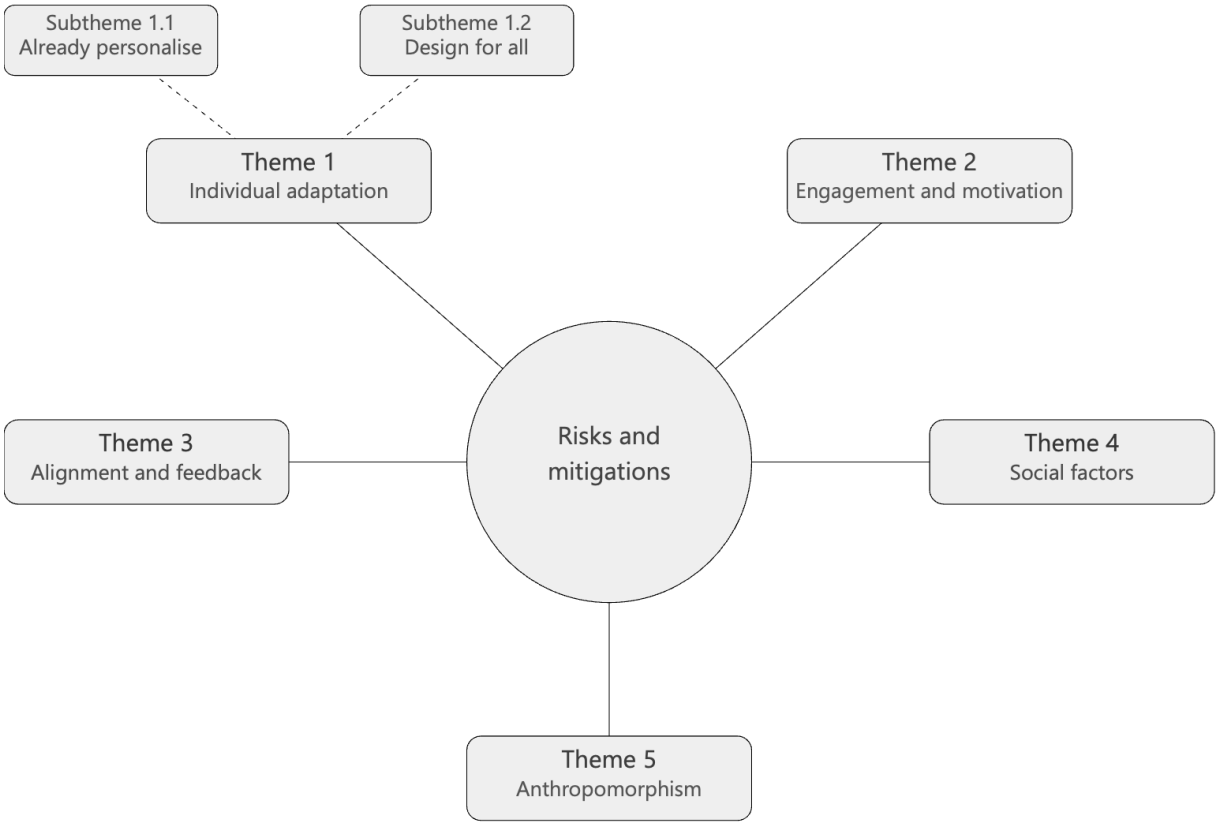


Figure 2: Thematic analysis summary

Theme 1: Individual adaptations are important

This theme captures a shared concern among the experts that overly generalised interventions risk being ineffective or inappropriate. Underlying this concern is an assumption that procrastination can arise from many different varying causes, and following this is a preference for tailored rather than standardised approaches. It highlights the experts' views that adaptations to the individual are crucial in mitigating these risks. This was the theme that occurred most often in the dataset across all three interviews. One excerpt that clearly shows the central point here is:

> *I think the personalisartion part of it is important, so it feels personal, it feels unique to me. I have no idea how to design it. But I think the person needs to feel that if I respond to questions from the chatbot, it will be used in some way to tailor the next step of the treatment or the modules to my specific problems. It does not feel too generic. I think that is most important. – P1*

Note how the expert has *"no idea how to design it*", this points to the complexity of the problem. This emphasis suggests that perceived individual relevance may be central to user engagement, not merely the technical accuracy of the intervention. It is not just about feeling personal and unique but also providing the correct type of treatment based on individual challenges.

> *And if you have a very generic type of treatment, you will miss out on that. And then you might provide the same treatment for everyone, even though the difficulties might be unique for the patient. I see that as a limitation. – P1*

Personalisation was not only described in terms of stable individual traits, but also as dependent on the individual's current situation and stage of development.

> *And it is supposed to be for personal development. It needs to have some knowledge about where this person is in its development, where it is in its status, its current situation. If it is about procrastination, it needs to have knowledge about this person. – P3*

This highlights that while procrastination creates a need for tailoring to individuals, other areas might not. There were also points being made about how psychiatric disorders can change the required treatment type altogether.

> *When I see patients face-to-face with problems with procrastination, I am always able to do an assessment of their individual difficulties. So sometimes that means identifying other psychiatric disorders that might lead to procrastination, such as depression or ADHD. And then I would give them another type of treatment than treatment for procrastination. – P1*

So, in some cases minor tweaks or adaptations might not suffice, and a completely different methodology can be required.

Subtheme 1.1: Chatbots already personalise

While personalisation is framed as essential, experts also point out that recent developments in chatbots already provide a basic form of personalisation. This introduces tension between perceived limitations and existing capabilities.

> *A form where you can ask questions, you get answers, you can ask follow-up questions, you can ask clarification questions, you can ask the model to contextualize the general advice it might have to your specific situation. This is something that was just not possible before. – P2*

This suggests that the problem has already been partly solved. The worry might be more concerned with the extent of these capabilities. One of the experts suggests an exemplary solution.

> *But there might be ways around it. I mean, if you have a chatbot, for example, then you might be able to ask questions that probe for these different issues and tailor the treatment accordingly. – P3*

Whilst the example illustrates one potential solution, the repeated use of the words 'might' signals uncertainty which emphasises the concern of if the capabilities are enough.

Subtheme 1.2: Design for all

A related perspective challenges the assumption that personalisation must always be specific to the individual. One expert introduces the concept of design for all, which entails including design for edge cases like mental disabilities in the general design for all to benefit.

> *It could be. But if you are thinking about design for all, you are trying to design for everyone. The idea is that you are designing for people with some kinds of disabilities, which then also benefits others. – P3*

This might at first seem to stand as direct opposition to personalisation, but the idea is that some adaptations are helpful to most or all users.

> *But then also if you look not only at an app, but at this entire service, there are different ways of conceptualizing what it means to do inclusive design. You can do it in a way that involves people with disabilities and design for them, which will benefit all. – P3*

The result is then therefore not a generalised design for the average user, but a design for all.

Theme 2: Users must be engaged and motivated for treatment to work

This theme captures the view that a minimum level of user engagement and motivation is a necessary condition for effective treatment. Across interviews, experts highlight both factors that may undermine engagement in chatbot-based interventions and mechanisms through which it may be enhanced.

> *Well, it mimics how you would do it face to face, but the difference is you have to put in more effort yourself as compared to face-to-face therapy. And it is also something about going to some place physically that might work better for some people when you see a therapist face to face. It might help you motivate yourself to do the homework assignments that are included in therapy*

> *compared to when you do all the things yourself via your computer or your smartphone. – P1*

Here we see one example of a potential motivational issue when using a chatbot as compared to seeing a real therapist. The specific case suggests not all risks can be avoided through design choices, as some stem from the intrinsic characteristics of a chatbot (here not being a person that one can engage with face-to-face).

> *And I guess the largest potential is in the fact that it is a much more engaging way to find information, but also to reflect on the answers that you get than what you would have with a search engine, say, where everything basically is on you, right? – P2*

Another expert considers the potential engagement of chatbots their largest potential benefit. Crucially, the reference point to which they compare chatbots are different, one compares it to other computer programs and another to traditional therapy. This comparison highlights that perceptions of chatbot effectiveness depend on the reference point used, whether traditional therapy or other digital tools.

> *But there are many kinds of engagement. You can be engaged in what you are doing intrinsically, and then you would need to have help in removing distractions. – P3*

Sometimes users might be interested in the task itself, reducing procrastination in our case. So we might not always need to increase engagement but rather decrease de-engagement like distractions.

> *There is a theory for motivation called self-determination theory. And research in motivation says that if you have external motivation factors, like rewards and things like that, when you gamify, you can get motivated to do it, but it is based on external things. But this can lead to superficial learning. Whereas if you are intrinsically interested in the thing that you are doing, that is when you get the real motivation, the real engagement, so to speak. And the factors that contribute to intrinsic motivation, according to this theory, to self-determination theory, is autonomy. That is, that you can choose for yourself what to do, what to focus on. Feeling of competence, that you actually achieve something. – P3*

The discussion of self-determination theory introduces a distinction between different types of motivation, particularly between externally driven and intrinsically motivated behaviour. The expert suggests that while techniques such as gamification may increase engagement, they risk promoting more superficial forms of motivation that do not translate into meaningful or sustained change.

Theme 3: Alignment requires feedback

This theme captures how experts frame alignment with human values as a central requirement for ensuring that chatbot-based interventions are both useful and safe. Human feedback is described as a key mechanism through which such alignment is achieved.

> *But these emergent abilities that we see with large language models, the fact that they can follow instructions, or they can answer questions, the fact that they not only produce grammatical language, but actually useful language or purposeful language, that is the outcome of the later stages of training that you use for ChatGPT, for example, which are called instruction fine-tuning and alignment with human feedback. – P2*

This highlights that the usefulness of modern chatbots is not solely a result of their base training, but of subsequent alignment processes guided by human input. In this view, human feedback is not an optional addition, but a foundational component in shaping meaningful and goal-directed behaviour.

> *It is not a human; it is just a computer program. But if we want to say that anyway, the model does not know if something is right or wrong. Nothing in the pre-training stage is giving the model that kind of signal. – P2*

The absence of such alignment is framed as a potential risk. Without signals guiding what is correct or appropriate, the model is described as lacking any inherent preference for truthfulness or helpfulness. This suggests that unaligned systems may produce outputs that appear plausible but are misleading or harmful.

> *And then you send that feedback all the way into the weights of the neural network to update them, so that the model is no longer just a next word prediction model, but a model that is aligned with what humans consider to be useful or perhaps ethical. That is another purpose of alignment that you want the model to produce outputs that are, you know, that comply with basic ethics, for example. – P2*

Alignment is further described not only in terms of usefulness, but also in relation to ethical considerations. Human feedback is seen as a way of embedding normative constraints into the system, guiding it towards outputs that are not only effective but also aligned with human values.

Theme 4: Social factors shape outcomes
This theme captures how interpersonal relations can affect treatment outcomes.

> *Well, so internet-based treatments overall, when it comes to procrastination might be problematic for some because you do not see someone face-to-face. You are not accountable to someone face-to-face. And if you have problems with procrastination, you might just drop out of treatment. there is something about this physical contact that helps some people. – P1*

Even if the treatment is mediated by a human, the physical aspect of sociality can play an important role for how well it works. This indirectly suggests a risk with chatbot treatment, which not only lacks the physical aspect, but the social aspect altogether.

> *And that is what we also seen in one trial where we compared a group treatment to internet-based self-help treatment. A group treatment was more effective in the long run. And we think it has to do with this accountability factor. You actually have to show up and say what you have been doing since last week. – P1*

The feeling of accountability does however seem to depend on more factors than if treatment is physical or digital. Group treatment may be difficult to replicate in chatbot-based systems, and this subsequently presents a direct challenge for the method. Advantages from group treatment seems to extend to other areas as well.

> *And the third thing that drives intrinsic motivation is, doing things together, basically. Motivation from others. Yeah, and from the sense of belonging together, and the sense of being part of a family or community. – P2*

Without the comfort of not being alone in one's struggles, there's a risk people feel discouraged by the thought that they are uniquely challenged, or lonely in their struggles. Even if a chatbot could present statistics on what has worked for others, explain how common the person's problems are, and simulate understanding, this suggests that there is something unique about talking to real humans. But this unique quality is not necessarily limited to benefits.

> *For some patients, not seeing someone face to face or not seeing a therapist at all could actually be a positive thing because they might have problems with the interpersonal relationship that will prevent them from benefiting from treatment. -P1*

Human connection can for some introduce elements of for example guilt, shame or stress. Even without negative-valence emotions, social interaction can add a layer of complexity that for some might obscure practical advice or exercises. Adhering to social norms during the treatment, for example, is something people might feel they have to think about on top of the treatment itself.

> *But overall, when you look at the results from self-help treatments delivered via the internet, we do not see any difference to face to face treatments. So essentially the benefits should be the same. It is just that the preference might be different between different people. – P1*

Not having the social dimension is then not necessarily a downside of chatbot-treatment but still presents a risk for a subset of the users. This again demonstrates the importance of personalisartion.

Theme 5: Anthropomorphism as a double-edged sword
This theme explores how the human-like behaviour of modern chatbot technology can result in both advantages and disadvantages. As chatbots come closer to mimicking human behaviour and language, it gets harder to differentiate between the two.

> *I am not sure whether I would say that this comes from how they have been developed. But if you think about it, how you use them, I mean, the great attraction of these models is that they behave in some way like a human. And we have this bias as humans that we interpret someone or something that behaves human-like as if it was a human. – P2*

Regarding procrastination treatment in particular, chatbots are an interesting new candidate treatment method in many ways because they are able to retain some aspects only humans could previously provide. Conversely, they are also interesting because they are not human; they are more readily available, require no salary et cetera.

> *We treat even computer programs, even chatbots that help us with booking a journey or whatever, as if they were humans. We feel compassion with them. We want to be polite. We take it seriously, what they are saying and so on. And of course, this is both a big benefit because it is a very natural way to communicate in this human way, to communicate with the model, but also a big weakness because they are not really humans, these models. – P2*

In anthropomorphising these computer programs, our behaviour towards them can change. The expert frames it as problematic precisely because they are in fact not human. This introduces a risk of misalignment between user expectations and the system's actual capabilities. Ideas from the previous theme can be used to illustrate how this could be both good and bad. If treating them as humans makes us feel more responsible for example, that could be a good thing. If it instead causes social anxiety, that would be undesirable.

> *It is easy to get into the fallacy that we are talking to a human and that we should take everything seriously, that we should feel bad if we say something weird or that we are not meeting a communicative goal or whatever. – P2*

What the expert is implying here is that we should in fact not take everything a chatbot says seriously and not feel bad if we say something weird to it. This in extension signifies that they are not yet fully trustworthy, nor do they have significant moral value.

> *If you talk to an AI assistant, then you may take their response, because you are interpreting this as the response of a human, as something that is true. You want to trust other humans, that is kind of a basic precondition of all forms of human communication. But because this is not a human, it is very easy to place false trust in the text, especially if it is so confidently presented as ChatGPT makes it. – P2*

We would not trust these chatbots if they could not resemble humans to the degree they can. ChatGPT's confident presentation is seen as contributing to this problem, this suggests that presentation style may influence the degree of trust users place in chatbot outputs.

5. Discussion

This section addresses the research questions in two stages: first examining potential risks of using current general-purpose chatbots for procrastination reduction, then discussing preliminary design implications for purpose-built tools. After this study limitations are discussed, and finally suggestions for future research are presented.

RQ1: Risks of using chatbots for procrastination reduction

This section means to address the first research question *"What are some potential risks associated with using generative artificial intelligence in the form of chatbots for reducing procrastination?"*, through discussion of the findings of the thematic analysis, and how they relate to the papers' theory. The risks presented should be seen as potential and preliminary, due to the exploratory nature of the study.

To start off, although the experts do not always directly frame generalised treatment as a risk inherent to the technology, the importance of individual adaptation is seen repeatedly in the dataset. We see this both explicitly in direct statements, and implicitly through understanding the implications of the other themes. Many potential issues seem to stem from an underlying lack of adaptability in treatment; over-generalization is a risk. Something crucial to keep in mind is that the importance of personalisation deeply extends into the other risk areas as well. This points both to the seeming interconnectedness of these topics and to the central role personalisation seems to play in answering the research questions.

Moving on, consider the risk of users not being engaged and motivated enough to follow through on treatment. In TMT (Steel & König, 2006) motivation is central in explaining why people procrastinate, the relevance of engagement and motivation is subsequently highly relevant as a risk. If users are not motivated enough then they will procrastinate, is in essence what the theory suggests. Either that means procrastinating treatment or it means continuing the procrastination they need help with in the first place. There is also the possibility of lowering the amount of effort put into the treatment, even if it continues.

Some tools that are normally used to increase motivation (including many social factors) seem difficult to implement into chatbots specifically. Motivation and engagement need to be created with the help of words and language, as that is the only method a chatbot has in terms of interaction. Some established motivation/engagement-increasing techniques might not be possible to implement at all, some directly applicable and others might need adaptation or reframing to fit into the chatbot context. Gamification (turning tasks into games) for example, on first thought might not seem possible, but could be implemented in terms of game played with language. Whilst more motivation generally is considered better, users can reach engagement levels higher than what is desired. Users might prefer spending time interacting with the chatbot instead of actually doing the thing that they are procrastinating. It can also be the case that the engagement-increasing activities distract users from the treatment itself, which is of course not desirable.

Another potential risk which emerges in the data is that of misalignment. Alignment training is done through the process of fine-tuning as described in the background section (Ouyang et al., 2022). Essentially this means that models can be specifically adapted to tailor to specific purposes (see background), and this means that advanced models with already a large number of parameters can be adjusted by relatively accessible means, there is no need to start from scratch. It is important to not view this fine-tuning as just another layer simply added on top of the original model. What is considered aligned behaviour in a general chatbot might be different (or even directly opposing) from what is needed in a specialised chatbot. A generally

aligned chatbot might be trained to be validating and emotionally supportive, whereas a procrastination-reduction chatbot might sometimes need to be challenging or confrontational.

It might also be a good idea to grow an understanding of what is not the goal. For illustrative purposes, imagine that a chatbot is successful in reducing procrastination, whilst at the same time increasing other undesirable behaviour or causing emotional distress. Even though that was not a part of what the chatbot was instructed to do, negative behaviour can arise as a side effect. Dealing with this problem directly could be challenging as it is very difficult to beforehand list every desired and undesired outcome (the possibilities are virtually endless) (Amodei et al., 2016).

From examining the existing literature, one can see that alignment in general seems to be one of the central components of chatbot-design, and the interviews suggest that this holds true for procrastination reduction as well. One dilemma is that what answers users like or prefer might not be what is clinically most effective. They might for example prefer a gentler and kinder version of the chatbot which helps them in rationalizing undesired behaviour, which then leads to sycophancy as described earlier (Sharma et al., 2024).

Another issue is that there are also differences in views amongst humans. We, as cultures and as individuals, have differing values, opinions and goals. Considering this is important, as the researcher's idea of what the chatbot should accomplish might differ from what users themselves want of the chatbot, maybe they truly prefer the kind over the effective for example. With a large dataset and feedback from many users, average preferences would emerge. But this also raises the question of whether the chatbots should be designed for user wants or if it should be designed with a specific task/goal in mind from the start.

There are also potential risks associated with social factors, and it might not seem obvious at first that they are an important part of how a computer program can aid us in procrastination reduction. While chatbots can elicit social *responses* in users (Nass et al., 1994), there is a meaningful question of whether the interaction constitutes genuine social engagement in the fuller sense, one that has historically been understood as requiring mutual conscious recognition. This question is relevant for this discussion for two reasons.

Firstly, it is *because* chatbots are not social beings themselves that we need to consider what is being missed out on, or potentially; what is gained. Psychiatric treatment face to face is fundamentally social, and from this stem both advantages and disadvantages (Flückiger et al., 2018). Secondly, chatbots are designed to mimic human behaviour. This can lead to humans treating these chatbots as social beings (to varying degrees). In other words, even though the social aspect of the chatbot itself is an illusion, that illusion still has real effects on the interaction. So, the social aspects matter because chatbots are not real social beings, and because they still elicit social responses from humans. These viewpoints might be seen as incompatible, but they each capture aspects the other misses.

Another important point is the individual variation of social behaviour. For one person social accountability might contribute to increased motivation, for another it might worsen the problem. Feelings of accountability towards others can be helpful in general when we want to increase the likelihood of actually doing something we want to do. As seen earlier, procrastination is by definition the unreasonable delay of action. The reason why accountability is helpful in general is then directly related to the core issue procrastinators have. As a result, the removal of social accountability stands as one of the more central potential risks of chatbot intervention for procrastination.

Just as these chatbots emulate social behaviour of humans, they do this with our whole written language. Humans function in large parts through language, it is our main way of communicating with each other, our prime way of expressing ourselves and developing ideas. It is deeply engrained in our identities and cultures. It should not be surprising then that if a chatbot is successful in producing convincing human-like language, our anthropomorphic tendencies activate, see the background (Epley et al., 2007) for more information. There are benefits to this, and in some ways, it is the point of making these programs to begin with. Artificial Intelligence is created to be able to do things that normally only humans can do, that is the very idea of it (Russell & Norvig, 2020). We create these programs because they can do things that "normal" computer programs cannot.

One major issue is that the models are working on a level exclusively based on language. There are, by design, no thoughts behind the language. Any resemblance of reason is simply a byproduct of imitating human logic and reasoning. Complex behaviour can nevertheless emerge from this linguistic process, as one expert mentioned. But still, when we anthropomorphise, we risk misleading ourselves that the level of sophistication in language of a chatbot is somewhat representative of the intelligence which is producing it, because that is how it tends to work when it comes to humans. This might then lead to us overestimating or over trusting what the chatbot says. Not only because the arguments are convincing, but because we think that the presentation itself represents the quality of the content, while that might not be the case at all, consider for example the tendencies AI has to hallucinate (Ji et al., 2023).

RQ2: Mitigations of identified potential risks

The second research question of *"How could these potential risks be mitigated, both by users and developers?"* refers to the potential risks mentioned in the previous section. Here mitigations and design suggestions are presented. Consider these as early suggestions of potential designs.

An alternative perspective to that of seeing over-generalization as a potential risk is that many problems seem to be solvable by adaptations to the individual. As it is a rather multi-faceted concept, it can be helpful to break down the concept of personalisartion into its various aspects. By doing that we can compartmentalise the deep dives, and free ourselves of having to address the whole idea at once. The following two questions capture the relevant discussion points well:

- Which individual differences should change which chatbot behaviour?
- By which methods can a chatbot capture the relevant information?

The relevant individual differences are those which change users' responses in ways that affect treatment outcomes. Following on the logic of TMT (Steel & König, 2006), the relevant aspects are whether a difference affects expectancy, value, impulsiveness or delay. SDT (Deci & Ryan, 2000) provides the factors of autonomy, competence and relatedness to evaluate against as well. Furthermore, differences are not either relevant or irrelevant, but all exist on a spectrum. The useful question is rather if a difference is relevant enough to address, which can be answered through established motivational theories. The natural follow-up question is how the relevant differences are identified. Some important ones can be seen in the answers of the

experts. Mental disabilities, social skills and which stage of development a user is on are examples of individual differences that can be of great importance. To find other examples, one potential method of generating useful material is further interviews with both users and experts. For an example of how this could be done, see the 'Future Research' headline.

When a difference then has been identified as relevant, one needs to address the matter of how the chatbot ascertains information about an individual needed to make helpful adaptations. Some questions are trivial to ask, such as if the user expects a reward for the task they are procrastinating. Other questions might be such that the fact of the matter might be clear, but that the topic is sensitive and therefore requires finesse, for example when it comes to mental diagnoses. Other information might not be so simple to gather, such as to what degree the person is affected by social pressure. Some aspects of personalities can either be sensitive, or simply hard to properly self-evaluate (Nisbett & Wilson, 1977). For such cases an alternative ascertetion-method could be building a gradual understanding of the user over time, keeping something like a model to represent the user. User modelling is already an established concept within human computer interaction, Fischer (2001) for example describes user models as 'models that systems have of users that reside inside a computational environment'. To not include superfluous and irrelevant information, there could be predefined traits to monitor. As the user spends more time interacting with the chatbot the accuracy of the model would increase. In regard to this, developers have to strike a balance between initial accuracy (straight up questioning) and getting started in a reasonable time (user modelling over time), especially since procrastinators in particular can be negatively affected by delay of rewards.

For some people, there might be some piece of information about themselves that carries high relevance to treatment but is rather unusual. In these cases, there might not be justification to ask questions about this trait to every user. But for successful treatment the chatbot would likely benefit from knowing about this. This could for example be done by asking users if there is anything else they can think of that the chatbot should know before starting treatment. This does not ensure that every relevant difference is captured, as users might not realize everything about them that could be of importance. In some cases, some aspects of a person could be so that chatbot treatment is a method that simply should not be used. This suggests that adaptations need not stay within the confines of the chatbot but might expand into recommending other methods. This is an important step because it could potentially pre-eliminate the risks of chatbot treatment for those that are least well fitted for it.

One of the experts describes the concept of 'design for all'. Holding this perspective can be useful in producing a more useful chatbot overall. Through user tests, or by observing results over time, some individual adaptations might be identified as so generally useful that they should be programmed into the standard behaviour of the chatbot. This suggests that personalisartion is not only useful in addressing individual differences, but also as a tool in identifying general improvements. As a result, personalisartion seems to be an important step in designing chatbots for procrastination reduction.

In regard to motivation, TMT can help us understand how to mitigate the associated risks. Expectancy could be increased by providing users with statistics of previously successful treatments, for example. Design could address impulsive aspects in users by providing short-term satisfaction in providing users with direct feedback on their progress, for example. A chatbot might not be able to actually decrease the actual delay of the thing the user is procrastinating, but it could help users re-frame the situation with sub-goals for example.

We can draw from SDT (Deci & Ryan, 2000) to realise that we should try to facilitate for intrinsic rather than extrinsic motivation. Both can be used together but if we could somehow induce strong intrinsic motivation in users, they would likely not need external motivation as well. The seeming paradox here is: how can we externally create something that has an internal origin? The answer is that it can be done indirectly, by nudging users to think about why they might be motivated to do something. From that genuine interest or curiosity could arise. An adjacent question is if chatbots can provide long-term reductions in procrastination or if users will become dependent on the chatbot to not procrastinate. Looking at the question through the lens of SDT suggests that if intrinsic motivation could somehow be induced in users, that would likely increase the probability that the treatment would have longer-reaching effects. Implementations such as reminders would function as extrinsic motivators and would likely not provide much benefit beyond the immediate intervention.

Moving on to the topic of alignment, the fact that the alignment process needs humans to put in work creates constraints on how extensive the process can be. To get good results, it could be helpful to gather a substantial amount of feedback. When resources are limited, this could be difficult to gather before a product launch. In such cases, developers would be presented with a choice. One option is to launch the service at an unfinished state and gather feedback over time from users. Doing it this way risks not providing helpful treatment and then wasting people's time. An even worse possible consequence is causing users harm, which is a very real possibility if the model is not aligned enough. The other option is to put more time into the alignment phase before launching the service. The problem with this, other than that time often being one of the limited resources mentioned earlier, is that people would be getting help later. Developers will at some point have to decide that they have passed the threshold for when a launch is reasonable, as more training can always be done. Aligning through human feedback might then offer an indirect way of organically growing the model's comprehension of what we want from it. In this way we would not have to computationally pre-program every desired and undesired outcome, the model could learn what those are indirectly over time.

The mentioned risk of loss of accountability could potentially be addressed from two different starting points. Either accountability can be re-created, simulated or replaced. Chatbots could

for example behave as if they were impressed or disappointed when a person did or did not follow through with whatever plan they had. The other attempt at a solution is to accept this flaw and divert users who need to feel accountable to alternative methods. This because for some people accountability can be a hindrance, an unnecessary layer of extra complexity. Perhaps every method of treatment need not try to help as many as possible. The optimal solution is probably a mix between the two perspectives, and yet again highlights the importance of personalisartion.

The point about group treatment presents an interesting challenge as it shows promising results within more traditional treatment. Although this poses a seeming challenge for chatbot treatment, in that it could be difficult to integrate, that does not mean it is impossible. One example of a possible solution is to have a traditional human group chat, where one of the members is a chatbot facilitating treatment through moderation and treatment provision. This could retain advantages of for example accountability. The difficulty here lies in actually co-ordinating the creation of the group chat. If there isn't some external agent like a medical professional inviting people to this group chat, it is hard seeing how they would get in contact in the first place. There is also the fact that individual adaptations in chatbot behaviour would have to be reduced. At best, chatbot behaviour could be adapted to group averages, which would be more specific than the general population. It is however also possible for medical professionals to group together people with similar profiles, that need similar adaptations. Regardless, there are obvious problems with this implementation, but the example still stands to show that group treatments are in fact possible. The difficulty is practical rather than fundamental.

Earlier the difficulties of ascertaining all relevant information were discussed. One possibility way of dealing with this difficulty is to give the user the option to choose between several different pre-defined AI-personalities. These could then be pre-thought-out to be designed to be especially effective for certain kinds of personalities. Instead of adapting on the fly to the user's behaviour, which might take time to develop properly, one could for example have a chatbot that has been observed to work extra well with people who have tendencies towards social anxiety. The chatbot could in theory still have capabilities for adaptation over time but would provide a stronger starting point closer to what is desired.

Considering the risks associated with anthropomorphism, one strength that general chatbots already have is that they do in fact have the ability to adapt their language and their level of mimicking human social behaviour. This could be kept in specialised chatbots as well. They can be instructed to do this accordingly to the needs of the individual in question. It seems to be preferable that this is done early in treatment together with the other personalisartion processes, since it can change what the best approach is altogether.

There are other potential ways in which to attempt at mitigating the risks relating to anthropomorphism. One possibility is making chatbots less human-like in their language, but that also somewhat works against what we are trying to achieve in creating artificial intelligence, we risk losing many of the benefits the human similarity also brings. For example, if the language is too different from how we communicate as humans, that could lead to humans having more difficulties understanding the chatbot. Another option is through visual or interface cues reminding users of the non-human nature of the chatbot, and that they can make mistakes that humans have a hard time noticing. This is something that is already often implemented in models. Yet another is educating users before interaction, which is perhaps something that could be done by medical personnel when prescribing chatbot treatment.

Limitations of the Study

Perhaps the most limiting factor of all is that only three interviews were conducted. If more interviews were conducted, surely more potential risks would emerge. The usefulness of each added interview is less than the previous one. It gets more and more likely that risks that emerge has already appeared in previous interviews. This phenomenon is explained through the qualitative concept of saturation, the point at which additional data collection no longer yields new themes (Braun & Clarke, 2021). Although three interviews fall below what is typically considered adequate for full saturation, the benefit per interview ratio is not linear. Repetition is however also its own strength, if a particular risk is brought up again and again that solidifies the legitimacy of it. Pertaining to the thematic analysis one could say it would lead to stronger themes. However, due to time and resource constraints more interviews would also lead to each interview having to be shorter and given less time in analysis. What follows is a weighted choice between depth and breadth. In hindsight a few more interviews would have been preferable, but not a lot more.

The number of experts also put constraints on how many research/expertise areas could be covered by the interviews. Even though the experts' area of expertise were chosen carefully and deliberately to capture the most relevant areas, many related areas were left out. For example, the perspective of a linguist (which is not specialised in language technology) or a neurologist could potentially have brought new and interesting perspectives. An alternative approach would be to only interview experts within the same research area. That would likely have added more depth and rigour to that area in particular but would also have required a

narrower research question. Since the specific area of risks with using chatbots for procrastination reduction is understudied, breadth was seen as more important than depth, hence the choice of not designing it this way. Another limitation stemming from the areas being understudied is that none of the experts had direct experience with using chatbots for procrastination reduction specifically. No such expert was found to be contacted. If such an expert existed, that would have been a strong addition.

The choice of thematic analysis coupled with the small sample size also brought limitations. Two risks with strong prevalence in the literature, hallucination and sycophancy were not present enough in the data to make themes out of. The literature would suggest that these are more central issues than many of the risks prevalent in the data. The study could have been designed differently to have a higher likelihood to capture these perspectives as well. Additional interview questions could have been formulated about risks with high prevalence in adjacent areas. The interviews were designed to facilitate free exploration from the experts, not validate preconceptions. Another limitation of the method of thematic analysis is that it is always in some sense subjective, both because analysis is based on interpretation, and because one has to decide which quotes are chosen as relevant.

The absence of data from users can also be seen as a limitation. Users hold a unique perspective in terms of what their needs and problems actually are. Although they might not be able to see the risks as clearly as the experts, risks could be extrapolated from analysing their perspective. Although this could have been a valuable addition it was omitted due to time constraints. Experts were deemed to have a higher likelihood of generating useful material.

The interviews were conducted in 2024, and language technology is a rapidly developing field, during these two years much has happened and changed. The questions, and the expert's answers were developed with the context and understanding of the time. It is difficult to say exactly how large the difference would be if they were done today. Readers are advised to keep the age of the interviews in mind and practice interpretative caution.

Suggestions for Future Research

There is sure to be value in replicating the study, for several reasons. Firstly, as mentioned in the previous section, the interviews were conducted two years ago, and the field of AI has developed during this time. Secondly, interviewing other experts from for example linguistics or neuroscience would make available their specific insights and could very well generate more risks. Thirdly, it could validate the results. Experts in other research areas could be interviewed, that would likely also yield unique contributions. The perspectives of users could also be gathered.

Beyond extending the scope of qualitative inquiry, future work could also move toward applied evaluation of purpose-built systems. The suggested designs such as personality presets, gradual user modelling or simulated accountability mechanisms could benefit from rigorous empirical testing and would move the work from theoretical to practical. Ideally, each design suggestion could be tested in its own study. This could be done for example by a controlled study with a control group who uses a generalised model like ChatGPT or Claude and a test group who uses the same model but fine-tuned with a specific design idea. Participants could be tasked to work within a context known to elicit procrastination, such as an extended academic project. Based on that multiple metrics could be compared between the groups. Lee et al. (2025) present multiple useful metrics which can be quantitatively compared, such as time management, stress and irrational procrastination scores. To further identify which implementations function well together the test group could use a chatbot fine-tuned for two

or more designs at the same time. Follow-up studies could be conducted to identify which implementations result in persistent improvements.

There is also the possibility of exploring areas adjacent to procrastination with the same methods. This paper primarily focused on risks, but benefits could also be more thoroughly explored. It is also relevant to note that research on chatbots in general is likely to decrease some of the identified risks, but it could also introduce new risks that originate in increased complexity.

6. Conclusion

For this study, the thematic analysis identified five overarching themes, each accompanied by associated potential risks and strategies for mitigating them. The most central theme is that of individual adaptation. Essentially, generalised treatment seems to pose a risk to effectiveness of said treatment in a myriad of ways. The second theme to emerge was that of engagement and motivation. The potential risk in this case is that chatbots could be designed in a way which does not motivate or engage users enough for treatment to work effectively. Alignment is the third theme of the paper. In general, not aligning through human feedback might risk immoral or dysfunctional AI. Of specific importance to procrastination treatment, what is aligned might differ between a general chatbot and a chatbot with a niched purpose. Following this is the theme of social factors, where the most central finding is that chatbot treatment potentially risks treatment not working effectively because of the lack of social accountability. The dual nature of some users benefitting from social engagement whilst others are hindered by it is also of importance. Lastly, anthropomorphism might pose a risk in that it can give users a false sense of confidence in what the chatbot says but can also stand as a benefit in preserving some benefits from human treatment.

Developers could act to mitigate these risks in many ways. They can implement systems that personalise various aspects of the chatbot, they can work on aligning the AI through human feedback to fine tune it for the purposes of procrastination, they can design the interface to deal with risks not manageable from within the conversation itself. Users could benefit by being a part of the personalisation process by proactively disclosing relevant information that might alter treatment and be aware of the limitations and not take everything the chatbot outputs as true before scrutiny. On a system level, and potentially one of the most critical countermeasures, is to decide when chatbot treatment is not appropriate. For users whose circumstances make them poorly suited to chatbot-based treatment, redirection could eliminate the worst outcomes.

Two other takeaways are worth mentioning. Some risks seem to be inherent to the technology itself, both for generalised and specialised implementations, and others are addressable through design. This distinction is important because if someone's personality is ill fit in regard to the inherent risks they could be steered away from this type of treatment. The final point, and perhaps the most central one, is that the importance of personalisartion seemingly stretches well beyond the theme itself. Tailoring chatbot behaviour to the individual recurs as a mitigation repeatedly within the other four themes and should be considered a candidate for cornerstone of treatment.

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Appendixes

Appendix A

LINKÖPINGS UNIVERSITET

2024-02-29
Linus Tullstedt
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Consent Form for Participation in Research

Title: Identifying and Mitigating Risks: The Role of Chatbots in Reducing Procrastination
Researcher: Linus Tullstedt, Undergraduate, lintu193@student.liu.se
Supervisor: Ruijing Ning, Postdoc, ruijing.ning@liu.se

Introduction
You are being invited to participate in a research study about the use of chatbots for reducing procrastination. This form provides you with information about the study and seeks your consent to participate.

Purpose of the Study
The purpose of this research is to explore expert opinions on the potential risks of using chatbots in personal development initiatives, specifically in the context of reducing procrastination.

What Does the Study Involve?
If you agree to participate, you will be asked to engage in a 30-minute interview with the researcher. The interview will be conducted via video call and will cover topics related to your expertise, as it pertains to relevant topics.

Confidentiality
Your participation in this study will remain confidential. All data will be anonymized. Data will be stored securely and only I will have access to it.

Voluntary Participation
Your participation in this study is entirely voluntary. You are free to withdraw at any time without penalty. You are also free to retract any segments of the interview.

Use of Research Findings
The findings from this study may be published in academic journals, presented at conferences, or used in educational materials. Identifiable information will be used only if you provide consent below.

Audio Recording
The interview's audio will be recorded solely to assist the researcher with transcription. It will not be published in any format and will be deleted once it has been transcribed.

Consent
- ☐ I have read and understood the information provided above.
- ☐ I have had the opportunity to ask questions and have had them answered satisfactorily.
- ☐ I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason and without my legal rights being affected.
- ☐ I agree to participate in this research study.
- ☐ I consent to the audio of the interview being recorded.

Signature: _____
Print Name: _____
Date: _____
Place: _____

Appendix B

Interview 1	Interview 2	Interview 3
Could we begin by exploring the psychology behind procrastination? How do misconceptions about it affect our understanding and approach to treatment?	Could you briefly share your professional journey in language technology?	How do general human-computer interactions, human-computer interaction principles apply to the development of interactive tools such as chatbots aimed at personal development?
In your view, which treatment methods have shown the most promise in managing procrastination, and how can technology enhance these methods?	From a high-level perspective, what are your thoughts on the potential of large language models in enhancing personal productivity and reducing, for example, procrastination?	If you try to transfer the general interaction principles that might apply to chatbots as well, what would those be?
Would there be any significant differences between having the treatment on the internet compared to having it in real life?	So, what recent developments, within the sphere of AI and language technology, have made this possible now?	What are the most, to you, important parts of designing interactive technology? And which one is the most important, or perhaps the few most important ones? And which one of these might apply to this context?
What opportunities do you see in the broader application of technology within psychological interventions, and particularly for procrastination as well?	That leads me quite well into the next question. If you could explain the basics of how these language models work, I guess if you want to add anything on the explanation you just gave?	In designing interactive tools, what strategies can be employed to enhance user engagement without distracting the user?
To contrast that, could you identify any potential risks associated with employing these technologies, perhaps especially chatbots instead of a therapist when it comes to procrastination?	So, in these conversations with the current technology, what do you see as the biggest strengths of these systems and perhaps the largest weaknesses as well? What are they good at and what are they bad at?	Just so that I understand the part about gamification correctly, there is a risk that if you gamify, the user might interact too much with the game, rather than the purpose of the thing itself.
How can we ensure that these technical interventions are designed to respect and accommodate individual differences in how and why these problems are arising?	Can you identify any other risks or ethical concerns based on how these models have been developed technically?	How do you avoid that in the gamification implementation?

So, for example, if someone is depressed, they might need another kind of treatment than someone who has ADHD?	Do you see any risks when it comes to privacy? Is that anything to worry about, or are these models simply too large for that to matter?	From a human-computer interaction standpoint, what are the key considerations for ensuring that interactive tools are accessible and inclusive to a broad spectrum of users?
Could you elaborate on the importance of human oversight when using digital tools in therapeutic practices?	How do you see the field and the technology evolving to mitigate these risks? What are the most crucial steps to take?	So, it is preferred that you design for all rather than make additional adaptations for those that need them especially?
Do you have any advice for either users of these chatbots or the developers that would perhaps lead to better help? Perhaps you have not done research on chatbots specifically, but is there anything that you could think would be helpful for people that are working with these things to consider?	Would you say that developers could include these constitutional rights now, or do they have to wait on the research to catch up? Is there anything that developers can implement right now that helps with this?	That is important in general, right, feedback from users?
Is there anything else that you would like to share or emphasize that we have not covered regarding the intersection of these topics?	Are there any other thoughts or insights that you would like to share on the topic of using AI to tackle procrastination that we have not covered in this interview?	How can human-computer interaction principles guide the creation of technology that adapts over time to meet changing user needs and personal development goals? Because these goals are not statical, personal development goals for example are always changing.
I have one final question. Do you think that missing out on human connection could be an issue as well, perhaps not only for procrastination, but in general replacing therapy with chatbot communication. Could that human connection element be an important loss?	No further questions were asked.	How do you generally gather that information, how it is usually done?
No further questions were asked.	No further questions were asked.	So, all these adaptations are perhaps a bit hard to translate into software. Because the mesh… Or

		perhaps not, are there ways of adapting to the situation without access to eye tracking sensors and other monitors of the physical world that the user is in?
No further questions were asked.	No further questions were asked.	I do not know if you can answer this one, but even outside the specific realm of chatbots, what risks should be considered when designing technology aimed at behaviour change or personal development?
No further questions were asked.	No further questions were asked.	How do you design for meaningfulness? Like, generally, what causes people to drop out when using technology?
No further questions were asked.	No further questions were asked.	So, one large potential risk then is if these technologies are underdeveloped but still put out to market and used by people for these purposes, then that could lead to the wrong behaviour change?

Appendix C

Interview 1 – Cleaned Transcript

I = Interviewer, P1 = Participant 1

I:

Could we begin by exploring the psychology behind procrastination? How do misconceptions about it affect our understanding and approach to treatment?

P1:

The fundamentals of procrastination is that it is a self-regulatory failure, which means that people have intentions to commit to certain actions, such as fulfilling a goal or achieving a certain outcome, but they fail in doing so by engaging in another competitive activity. Procrastination is basically that you have difficulties engaging in goal-directed behaviours.

And the reason for why we do that is, well, multifactorial, but to a great extent it is related to the fact that we engage in more immediate gratifications instead of the long-term gratifications, that we have a hard time overcoming certain emotions that arise when we try to pursue the activities that we are supposed to pursue.

You said something about the misconceptions about procrastination. I think a lot of people might have an idea that procrastination is because they are lazy or that they do not have the character to do the things that they are supposed to do. They blame themselves for not being able to engage in the things that they are supposed to be engaging.

I:

In your view, which treatment methods have shown the most promise in managing procrastination, and how can technology enhance these methods?

P1:

So far cognitive behavioural therapy has been most promising. That is a form of psychological treatment where you try to understand what is maintaining this behavioural problem. It is not about what might have caused it to begin with, but much more how can we prevent it from continuing.

In terms of procrastination, cognitive behavioural therapy is focused on what is preventing us from engaging in the activities that we are supposed to engage in. And that could be being in the situation, even though you are experiencing negative emotions.

One component of treatment is exposure therapy. Exposing yourself to emotions that you find negative, such as anxiety or stress. Another component is to eliminate all the stimuli that otherwise might make you engage in competing activities. Removing all the distractions. There are other treatment alternatives, but most of them have some kind of connection to or relationship to cognitive behavioural therapy.

And you were saying something about internet and doing it, this type of treatment via digital means. Since CBT is very structured as a treatment, it is also very easy to transfer into another medium. you do not have to see a therapist face to face. You could deliver CBT as self-help treatment. The person reads up on his or her problems, engages in different homework assignments, try out new behaviours in real life, and then submits his or her results to a therapist online. it is a type of treatment that works very well via the internet.

I:

Would there be any significant differences between having the treatment on the internet compared to having it in real life?

P1:

Well, it mimics how you would do it face to face, but the difference is you have to put in more effort yourself as compared to face-to-face therapy. And it is also something about going to some place physically that might work better for some people when you see a therapist face to face. It might help you motivate yourself to do the homework assignments that are included in therapy compared to when you do all the things yourself via your computer or your smartphone.

I:

What opportunities do you see in the broader application of technology within psychological interventions, and particularly for procrastination as well?

P1:

In general, it allows us to disseminate evidence-based treatments to more people. The main advantage of using internet-based CBT, for instance, is that we have already tested out these methods face-to-face and we have seen the positive results. But since most people might not be able to access this type of treatment because they live in geographically remote areas or they do not have the time to go to a hospital, internet-based treatments would allow more people to get access to these evidence-based treatments. So that is a main advantage.

And it is more cost effective as well. One therapist can see more patients online because it is an asynchronous written communication instead of seeing someone face-to-face, which would require you to sit down with one patient per session. we would be more cost effective in the long run.

This is in general, but for procrastination, again, it would allow more people to get access to the treatments that work because procrastination is not a mental health issue. It is not a psychiatric disorder. you might not get any help if you go to, let us say the hospital and say, I have problems with procrastination. But if we could deliver this treatment via the internet, more people would get the right type of help.

I:

To contrast that, could you identify any potential risks associated with employing these technologies, perhaps especially chatbots instead of a therapist when it comes to procrastination?

P1:

Well, so internet-based treatments overall, when it comes to procrastination might be problematic for some because you do not see someone face-to-face. you are not accountable to someone face-to-face. And if you have problems with procrastination, you might just drop out of treatment. there is something about this physical contact that helps some people.

And then I think some other disadvantages could be that an internet-based format or other digital tools might not be engaging enough. People find it boring or not so motivating, they just drop out of treatment. And I think for procrastination more specifically, patients want to see results quickly. And this also relates to some of their problems because they are much into seeing the results of their actions. They are into the more immediate gratifications. If they cannot see that they are achieving something within the program, then they might drop out.

I:

How can we ensure that these technical interventions are designed to respect and accommodate individual differences in how and why these problems are arising?

P1:

That is a difficult part. When I see patients face-to-face with problems with procrastination, I am always able to do an assessment of their individual difficulties. So sometimes that means identifying other psychiatric disorders that might lead to procrastination, such as depression or ADHD. And then I would give them another type of treatment than treatment for procrastination.

But another part of it is also trying to configure or conceptualize their problems with procrastination, which is something you do ideographically. If you have problems with procrastination, that might be different from my problems with procrastination. You have to figure out that together by me asking questions and doing a so-called case conceptualization.

This is a central part to cognitive behavioural therapy that we use in this type of treatment. And that might be problematic if you have just one generic treatment that you give to everyone receiving treatment for procrastination.

I:

So, for example, if someone is depressed, they might need another kind of treatment than someone who has ADHD?

P1:

That is one part of it. You have to figure out, is it really procrastination that is causing the problems or is it something else? Let us say that you suffer from procrastination. You also have to find out what is maintaining the problems. Is it that you are procrastinating because you do not see the value in doing the activities that you are supposed to do? Or are you procrastinating because you are easily distracted? What are the driving factors here? And that you need to discuss together with the patient before engaging in any treatment.

I

Could you elaborate on the importance of human oversight when using digital tools in therapeutic practices?

P1:

When we deliver CBT via the internet, there is usually one module or one part of the treatment in the beginning where the patient needs to think of what is the maintaining factors for my problems. And then you submit that assignment to the therapist who then gives you feedback. And this is part of tailoring the treatment to the individual problem. And if you have a very generic type of treatment, you will miss out on that. And then you might provide the same treatment for everyone, even though the difficulties might be unique for the patient. I see that as a limitation.

But there might be ways around it. I mean, if you have a chat bot, for example, then you might be able to ask questions that probe for these different issues and tailor the treatment accordingly.

I

Do you have any advice for either users of these chatbots or the developers that would perhaps lead to better help? Perhaps you have not done research on chatbots specifically, but is there anything that you could that you think would be helpful for people that are working with these things to consider?

P1:

I think the personalisartion part of it is important, so it feels personal, it feels unique to me. I have no idea how to design it. But I think the person needs to feel that if I respond to questions from the chatbot, it will be used in some way to tailor the next step of the treatment or the modules to my specific problems. It does not feel too generic. I think that is most important.

I

Is there anything else that you would like to share or emphasize that we have not covered regarding the intersection of these topics?

P1:

In terms of procrastination, it is also good to know that a lot of people seek help when they are facing a crisis. A lot of people who come to me as a psychologist or who sign up for our clinical trials, they usually want to receive help immediately with a specific assignment or specific activity that they have been procrastinating. And that is quite difficult to do something with because it is usually just before a deadline and it is probably not going to help that person. Receiving a treatment for procrastination always must be about *"how can I become better in the future?"*. I might fail at this assignment because it is already too late to get some help. But the help that I might receive from this treatment could help me in the future when I am encountering similar activities later.

I

I have one final question. Do you think that missing out on human connection could be an issue as well, perhaps not only for procrastination, but in general replacing therapy with chatbot communication. Could that human connection element be an important loss?

P1:

Both yes and no. For some patients, not seeing someone face to face or not seeing a therapist at all could actually be a positive thing because they might have problems with the interpersonal relationship that will prevent them from benefiting from treatment. Let us say that you are very focused on how this therapist would interpret me or view me, then you might be having problems

staying focused on what the treatment rationale. For those people, a self-help treatment, chatbots or just reading material could be more beneficial.

But there are also patients who would prefer seeing someone face to face because that might be their perception of what therapy is. If you provide something else, they might be less inclined to engage in treatment because they want to see someone face to face because that is what they have been seeing on television or whatever. So that might not become motivating for them. They want to see someone face to face because that is their perception of therapy in general.

But overall, when you look at the results from self-help treatments delivered via the internet, we do not see any difference to face to face treatments. So essentially the benefits should be the same. It is just that the preference might be different between different people.

Appendix D

Interview 2 – Cleaned Transcript

I = Interviewer, P2 = Participant 2

I:

From a high-level perspective, what are your thoughts on the potential of large language models in enhancing personal productivity and reducing, for example, procrastination?

P2:

In general, I think with large language models (or perhaps it is better to talk about AI assistance) one of the big potentials is that you can interact with something that is a bit similar to a search engine, but in a completely different way, in a dialogue. A form where you can ask questions, you get answers, you can ask follow-up questions, you can ask clarification questions, you can ask the model to contextualize the general advice it might have to your specific situation. This is something that was just not possible before.

I mean, that applies not only to personal productivity and procrastination, but to all kinds of knowledge questions. I want to know something about a specific area, or I want to get feedback on something that I think, that I have written, that I have done. And as soon as you can communicate in natural language about that, you can use this language model as a kind of coach, if you want, right?

I mean, there are many metaphors, I guess coach is one, which is perhaps closest to personal productivity. I mean, it could also be as a kind of writing coach, or as a mentor, or as a sparring partner. I mean, you can come up with all kinds of metaphors there. And I guess the largest potential is in the fact that it is a much more engaging way to find information, but also to reflect on the answers that you get than what you would have with a search engine, say, where everything basically is on you, right?

I mean, you just get this feedback in the form of pages, but you as a reader still need to sift through all these pages and find whether they really interest you. And here you get something that feels like a human response, even though it is not.

I:

So, what recent developments, within the sphere of AI and language technology, have made this possible now?

P2:

I think there are two main developments. One is what you could call upscaling. I mean, language models have been around for a very long time and have been trained on what you today would call small text datasets. And with upscaling, I mean, upscaling of data. So, large language models today are trained on essentially the whole internet, if you want. Many petabytes of data, which is a ridiculous amount. It is very large.

Also, upscaling in terms of computational power. The computers that we have today are much more powerful than the ones that we had only five years ago. And, the methods that are used by these computers, like deep learning, machine learning-based methods, they are much more powerful today than they were previously. That is one main difference, I think, that made this possible.

And another development that is crucial for large language models is that previously a language model was only trained on raw text, on what is known as the next word prediction task. So, you try to make the model predict the next word, and then you give it feedback in the form of whether that word that the model predicted was the next word in the text data that you are using for the training. And that is still a very important stage of training.

But these emergent abilities that we see with large language models, the fact that they can follow instructions, or they can answer questions, the fact that they not only produce grammatical language, but actually useful language or purposeful language, that is the outcome of the later stages of training that you use for ChatGPT, for example, which are called instruction fine-tuning and entment with human feedback. So, I think these two together, upscaling and the new training techniques, have made ChatGPT and other AI assistance possible.

I:

That leads me quite well into the next question. If you could explain the basics of how these language models work, I guess if you want to add anything on the explanation you just gave?

P2:

Typically you say that there are, you can conceptualize it as, there being four different training stages. The first one is this next word prediction training stage. The second one is what you call instruction fine-tuning.

So, there you also use next word prediction, but specifically on texts that are written as dialogues or instruction following texts, so that you are not only predicting the next word in a news article, right, but you are predicting the next word in a dialogue between a human user and the chatbot, but a purposeful dialogue where the model is supposed to do something or to answer a specific question.

And then you have the last two stages are what is known as reward modelling, where you are basically building a model that you give feedback to the AI assistant as to how useful its answers are. And then you send that feedback all the way into the weights of the neural network to update them, so that the model is no longer just a next word prediction model, but a model that is aligned with what humans consider to be useful or perhaps ethical. That is another purpose of alignment that you want the model to produce outputs that are, you know, that comply with basic ethics, for example.

I:

So, in these conversations with the current technology, what do you see as the biggest strengths of these systems and perhaps the largest weaknesses as well? What are they good at and what are they bad at?

P2:

One thing that they are very good at, based on how they are pre-trained, is to produce grammatical language. And not only grammatical language, but also high quality, fluent, typically English language, where you can also tune them to specific styles. You can say *"I want the output in the form of a colloquial conversation."*, *"I want the output in the form of an academic text."*, *"I want*

this to be written in the style of Shakespeare." or whatever. So, they are very good at aligning with different styles and types of text. That is a clear strength.

Then another strength is that they have this dialogue capacity that I talked about earlier. They can kind of react to what you said. And they are trained on a very large data set, which means that they implicitly have captured a lot of knowledge that is available in the form of text on the Internet. And in some sense, these are also the reasons for why ChatGPT and other AI systems have weaknesses.

There's also the weakness of hallucination or confabulation, as we also say. And that is because the model is very good at producing convincing text. And it is actually trained at that task, or that is one of the major training tasks that it has been going through. So, it does not "know", and I am putting that in quotation marks because the model does not know anything anyway. It is not a human; it is just a computer program. But if we want to say that anyway, the model does not know if something is right or wrong. Nothing in the pre-training stage is giving the model that kind of signal.

This can to some extent be changed in the later stages of the training; in the alignment stage, for example. But basically, if the model has not been specifically aligned to a certain task, there is always the risk that it will just produce convincing text. Because it has been trained to produce text without having any idea of whether that is truthful or not.

Another big limitation, just to mention it briefly, is that everything that is in the data is reflected in the model. And that is both a strength and a weakness. Because if you are training on internet data, for example, internet data gives you a skewed view of how humans behave. So, this will be a bias in the model.

And, finally, a big weakness, I think, is that most of the internet data is in English. Whereas many smaller languages are not reflected or do not have as much data. And therefore, the models will not be as good on those languages as they are in English.

I:

Can you identify any other risks or ethical concerns based on how these models have been developed technically?

P2:

I am not sure whether I would say that this comes from how they have been developed. But if you think about it, how you use them, I mean, the great attraction of these models is that they behave in some way like a human. And we have this bias as humans that we interpret someone or something that behaves human-like as if it was a human.

We treat even computer programs, even chatbots that help us with booking a journey or whatever, as if they were humans. We feel compassion with them. We want to be polite. We take it seriously, what they are saying and so on. And of course, this is both a big benefit because it is a very natural way to communicate in this human way, to communicate with the model, but also a big weakness because they are not really humans, these models. It is easy to get into the fallacy that we are talking to a human and that we should take everything seriously, that we should feel bad if we say something weird or that we are not meeting a communicative goal or whatever.

And in the context of information finding in particular, another big risk that I see is that for a search engine that we talked about earlier, you as the user still have an agency in sifting through these web pages that Google finds for you, for example, and deciding which ones are the ones that you find interesting or that are relevant to you. You are also looking at these web pages and ask yourself the question, can I trust this source? Because it is still something that is an artifact somewhere on the internet.

If you talk to an AI assistant, then you may take their response, because you are interpreting this as the response of a human, as something that is true. You want to trust other humans, that is kind of a basic precondition of all forms of human communication. But because this is not a human, it is very easy to place false trust in the text, especially if it is so confidently presented as ChatGPT makes it.

I:

Do you see any risks when it comes to privacy? Is that anything to worry about, or are these models simply too large for that to matter?

P2:

That is definitely something to worry about. Every time we give away data, we should worry about how it is used, and we should make sure that we are okay with the way in which, in particular in this case OpenAI for example, for ChatGPT, is using our data. There have been many demonstrations of adversarial attacks against language models, where the model is prompted in such a way that it reveals specific information from its training data. Even personal information, names, birthdates, everything it can find on the internet, and everything that we as users provide to OpenAI as feedback.

It is very hard to know whether that is used for training, or whether it can be used at a later stage, so we should be very critical of how our data is processed. Of course, in some sense, we should also be that for all kinds of other products that we are using, but it perhaps becomes more pertinent for language models.

I:

How do you see the field and the technology evolving to mitigate these risks? What are the most crucial steps to take?

P2:

That is a very good question. I am not sure whether there is a good answer to it. I can give you some examples of what people do. One example is, of course, to be very transparent about the data that you use for training these models, so that at least in principle, you can kind of explain how the model gets to a certain, or why the model produces a certain output. Because, of course, the model is not a magical device. It relies on training data.

So, if you observe a weird model output, for example, and you (at least in principle) know what the model has been trained on, then you can trace that output back and kind of explain it at that level at least. That is because it is in the training data. So, transparency.

Another thing that happens is that people try to, in this alignment phase that I talked about earlier, explicitly align the model with certain values. So, for example, with the Declaration of Human Rights (with European values) in terms of democracy. And then the interesting question is how to operationalize these values.

I mean, we know that we as a society have these values, and they have sometimes been codified as in the Declaration of Human Rights, but then the next question is, how do you operationalize that and put it back into the training procedure for the model?

This is also sometimes known as (a related term is) constitutional AI, where you are basically trying to define rules or guardrails that the model should use or should respect. And then build that into the systems as either, a watchguard or directly into the training procedure so that you perhaps exclude training data that violates certain standards. These are two techniques that are being used.

I:

Would you say that developers could include these constitutional rights now, or do they have to wait on the research to catch up? Is there anything that developers can implement right now that helps with this?

P2:

I would say that this is ongoing research. I mean, it is quite intuitive that based on how the models are trained, you should be able to influence the models. But basically, that is a research question. For example, we currently have an ongoing research project where we develop a large language model for European languages or Germanic languages. And that is one of our work packages that we want to explore. The use of techniques similar to the ones that I described, to make the model more trustworthy and more transparent.

I:

Are there any other thoughts or insights that you would like to share on the topic of using AI to tackle procrastination that we have not covered in this interview?

P2:

I am perhaps not so qualified to say specifically something about procrastination. Regardless, one thing that I always like to talk about in the context of language models or AI assistance is this term agency that I mentioned earlier.

I think it is very important that we understand that this is, (ChatGPT, for example) in the first place a technical artifact that we as humans are using. And as many technical artifacts, perhaps all technical artifacts, they can be used for good or bad things. And I think it is important that we understand that we have agency over that.

We are the ones to decide how to use the model. And we are the ones to decide also how to interpret what comes out from the model. And if you now place it in the context of procrastination, of course, it is kind of a tricky field because procrastination, to some extent, is about giving up agency, right?

To not take active decisions on, I want to do this now, but more like getting into this passive mode where you are delaying your own decisions and you are just, you know, I do not know, distracting yourself from the fact that you perhaps need to take a decision. So perhaps in this context, it is extra critical to remind everybody, every user of a language model about the fact that you still have agency over the model. And you can decide, for example, to not use the model, or if you use it; to be a responsible user and be aware of the limitations.

But also, of course, there is a lot of potential in these models. I mean, I do not want to say that you should not use language models, but I want to make it clear that, or from my perspective, at least, it is very important to remind ourselves about the fact that we are users that can decide how we use the models.

Appendix E

Interview 3 – Cleaned Transcript

I = Interviewer, P = Participant

I:

How do general human-computer interactions, human-computer interaction principles apply to the development of interactive tools such as chatbots aimed at personal development?

P3:

I really do not know, because I am not an expert in the field. So that is the first part. I would say that it depends on how you design the chatbot. There are, I mean, the general ones that have to do with things like interaction, not necessarily the things that has to do with the visual design, because the chatbot is not about the, it is not like a graphical user interface, it is an agent that you are interacting with. And there are, I mean, there are a bunch of other guidelines that probably are applicable. I have not designed that many chatbots, so I do not really know.

I:

If you try to transfer the general interaction principles that might apply to chatbots as well, what would those be?

P3:

I am trying to think about Nielsen's heuristics. And, of course, they do apply to some extent. Not all of them, not like, the aesthetics part. Altough perhaps. It would, if you extend it. It depends on how you interpret it. He has one about simplicity and simple aesthetics and not messing up the user interfaces. If we extend that into things like, well, you need to be able to stay on topic and not talk a lot of gibberish. So that would be equivalent to clutter on the user interface.

I:

What are the most, to you, important parts of designing interactive technology? And which one is the most important, or perhaps the few most important ones? And which one of these might apply to this context?

P3:

There are so many, and that is the problem. And that is why Nielsen and Mollick compiled their list of nine and then extended to ten rules of thumb. Based on reading huge amounts of papers, around a thousand of them, literature reviews, and then distilling them down into a few principles. And they were able to distil it down to ten.

This applied to user interfaces that were prevalent at that time in the beginning of the 90s, which means text-based user interfaces or command-based user interfaces, command-line based. And menu-based user interfaces and direct manipulation and graphical user interfaces. Well, menu-based also graphical. So those were the main kinds of user interfaces that existed at that time.

But there were also dialogue systems that you could have a dialogue with, what we today call chatbots, which is a simple form of dialogue systems. Because chatbots usually do not have very good memory of what you have been talking about, which more advanced dialogue systems do have. Okay, so they did have that. Yeah, well, they were around. There were such user interfaces. I mean, they have been around since ELISA. And then we are back at, when was ELISA, 71? Yeah, I think something like that. So they have been around.

But I think that there are different set of principles that apply for the chatbots and the dialogue systems. But as I said before, that is not my main area of research. So therefore, I cannot say what they are because I have not read up on it.

So, looking at these particular kinds of systems, then, I would say that it is managing a conversation that is important and how to manage a conversation. How to move it forward, how to stay on topic, how to drive it with some purpose. And all of those things are, I think, the most important thing for a successful system as the one that we are talking about now.

And it is supposed to be for personal development. It needs to have some knowledge about where this person is in its development, where it is in its status, its current situation. If it is about procrastination, it needs to have knowledge about this person. Is this person procrastinating right now or is it actually using me as a chatbot? Yeah, so it needs to have knowledge about the individual. Yeah, and knowledge about any system for self-improvement or behaviour change.

There is a researcher that writes about behaviour change, behaviour change support systems, as they are called. So where is this person now? Where would this person like to be when it comes to personal development? And how do I get this person to go in that direction? And where is this person... That is one kind of developmental path that you need to have understanding of.

Then you have the other developmental path that is, what state is this person in right now? And is this person in a procrastination state right now or is it in a productive state? If it is in a productive state, why the hell is it using me? So, I mean, yeah. Yeah, you would have to have some specific... The model would have to have some specific knowledge just about development and the person in question.

That needs to direct the conversation. And that is why you can not use the current large language models like ChatGPT, like of the shelf. Because you need to direct it. Yeah. Otherwise, there could be a risk that it does not really know what it is doing. Then it instead becomes used for procrastination instead of counteracting procrastination. That is what I see as the biggest risk. So, my answer to that question is I do not know.

I:

In designing interactive tools, what strategies can be employed to enhance user engagement without distracting the user?

P3:

In gamification, if we are using that, for instance, as a means for reaching engagement with like short-term goals and rewards and things like that, and feeling of accomplishment that you actually do accomplish something, and those kinds of principles that you have in gamification. If you do gamify things, it becomes important with alignment, goal alignment, so that you do not start playing the game instead of doing what you are supposed to do. You need to align the goals of the gamification with the goals of the activity that you are supposed to do. And in that way, if you manage to do that, you can use gamification to increase engagement.

But there are many kinds of engagement. You can be engaged in what you are doing intrinsically, and then you would need to have help in removing distractions. You know, when you have a full-screen mode in your word processor with a simplified user interface, so that it is only you and the text, that can, for instance, facilitate engagement with the text or the activity of writing. Just simply removing the distractions.

The same thing goes for reading. If it is only you, the text and nothing else. And you can get help in removing distractions, like you put on your noise-cancelling headphones. I have done studies of reading recently. My children and fourth graders are reading books. They can then put their hoodie on, and if you have long hair, you can use that also to shield off your attention.

You can sit like this and read, and you will not get distracted by people who are moving around. Removing all those distractions can get you immersed, in that case, into the text. So, that was gamification and then removing distractions. There is two obvious ways to do it.

I:

Just so that I understand the part about gamification correctly, there is a risk that if you gamify, the user might interact too much with the game, rather than the purpose of the thing itself.

P3:

Yeah, there is a risk of creating shallow interaction. So, you start playing the game instead of actually learning something, instead of doing something real.

I:

How do you avoid that in the gamification implementation?

P3:

I am not sure. And this has to do with motivation. There is a theory for motivation called self-determination theory. And research in motivation says that if you have external motivation factors, like rewards and things like that, when you gamify, you can get motivated to do it, but it is based on external things. But this can lead to superficial learning. Whereas if you are really intrinsically interested in the thing that you are doing, that is when you get the real motivation, the real engagement, so to speak. And the factors that contribute to intrinsic motivation, according to this theory, to self-determination theory, is autonomy.

That is, that you can choose for yourself what to do, what to focus on. Feeling of competence, that you actually achieve something. And here, the gamification involves balancing skill and difficulty, for instance, to reach a flow. Can not be too easy or too difficult, because if it is too easy, it gets boring. If it is too difficult, you get excited instead. So balancing difficulty and competency is probably one key thing, according to self-determination theory, and I think it is correct. And the third thing that drives intrinsic motivation is, doing things together, basically. Motivation from others. Yeah, and from the sense of belonging together, and the sense of being part of a family or community.

I:

From a human-computer interaction standpoint, what are the key considerations for ensuring that interactive tools are accessible and inclusive to a broad spectrum of users?

P3:

There are so many ways to cut the cake here. So, making sure that technology is accessible is, I mean, there are some basic hygiene factors, so to speak. And that has to do with, like, can people access the technology even with permanent or temporary impairments? And we can look at, I mean, like physical or sensory impairments for starters. That is the easiest part.

And that you have, like, the W3C guidelines for accessibility and things like that. There are a bunch of guidelines that you need to follow and make sure that your user interface or application is accessible, on phones as well. If you have a screen reader, for instance, it can be used with the phone's built in screen reader, covering the whole user interface and stuff like that. This would help if you have difficulty seeing, for instance. So those are the simple ones. But then when it comes to procrastination, there may also be cognitive disabilities involved. So we might, for instance, have ADD or ADHD involved.

I:

Perhaps it would be beneficial if the program knew that the person needed that kind of practice.

P3:

It could be. But if you are thinking about design for all, you are trying to design for everyone. The idea is that you are designing for people with some kinds of disabilities, which then also benefits others. So, I mean, I am going to give you an example.

I was part of a project that was about managing money, managing your finances when you have ADHD. And a problem with ADHD is that it has one component that has to do with impulse control and one component that has to do with, which may be related, working memory impairments or reduced working memory capacity.

And there is a connection there with attention as well, because working memory, you use that to hold attention. And one thing this leads to is that you tend to lose all your money before the end of the month because you spend it, because of the impulse control, but also because money or budget over time becomes a complex concept that is difficult to keep in your working memory when you have problems with your working memory.

What we did, or what they did, at this design firm, together with Attention, which is the national society for people with neuropsychiatric diagnosis, is that they did a visualization of your budget over time and how much you can spend and cannot spend. And where you put in: "I am going to buy this, can I do that?". And they said "well, yes, you can, but then this will be the consequence for the rest of the week if you do that". So that was like a feed-forward mechanism that was embedded into the application. So, design for all, or accessibility, can mean many different things. Especially when we move beyond the physical and sensory disabilities also into cognitive impairments of different sorts or disabilities.

I:

So, it is preferred that you design for all rather than make additional adaptions for those that need them especially?

P3:

Yes, that is the principle of inclusive design. But inclusive design also involves then working together with people with disabilities to figure these things out, which then directs how you set up the design process.

I:

That is important in general, right, feedback from users?

P3:

Yes, but also in whom you listen to and who you recruit. But then also if you look not only at an app, but at this entire service, there are different ways of conceptualizing what it means to do inclusive design. You can do it in a way that involves people with disabilities and design for them, which will benefit all. That is sort of traditional inclusive design perspective.

But we can also look at it from a systems perspective and say that, well, if we are delivering a service, we need to cater for a variation of different users. And this is like a more cybernetic view of service design. So, our service needs to be able to handle a variation of incoming users and still have them to go through our service. And then there is yet other approaches to it.

I:

How can human-computer interaction principles guide the creation of technology that adapts over time to meet changing user needs and personal development goals? Because these goals are not statical, personal development goals for example are always changing.

P3:

So, what you need to look at then are the field of adaptive user interfaces and intelligent user interfaces. And user modelling as part of that. That is the research fields. I have only sort of skimmed the area, so it is not really, again, not really my research area. I would have to read up on it to give any definite answers to it.

But if you look at an architecture for adaptive user interfaces, as I said before, it involves knowing things about the user and what their capabilities and how they are. We need to know something about the situation that they are in. And we need to know a bit about the task, what they are trying to do.

I:

How do you generally gather that information, how it is usually done?

P3:

It depends on context, what kind of system it is. So, when we have been looking at adaptive user interfaces for cockpits, for pilots, for instance, we need to look at what is their current workload. And then we can use eye tracking and heart rate variability and other sensors to look at what is the current workload and what are they focusing on (by eye tracking). We can also look at how are we currently presenting things for them. So, what is the situation like? Is it noisy? So, that was the user. And in that case, it is about the workload for the user that we are monitoring.

And for the situation, is it noisy? Is it stressful? Well, we can measure the stress level on the user also. Is there a lot of things going on visually? What kind of situation is it? Is it a time critical situation? Or is it a situation where you have lots of time? That might be part of the task modelling also. It is difficult to say if it is the context or the task.

If we, for instance, say that we are trying to achieve this, but now we have a really busy situation and then we might, for instance, the adaptive user interface can then say, well, since the user is really busy right now, and this is not time critical, given the task, that we know where we are in the task, we can wait to present this until later. Because right now, this pilot is trying not to be shot down. And that is a bit more important than for example managing some resource of some sort. And avoiding getting shot down is perhaps more important than even continuing towards the target or something like that.

I:

So, all these adaptions are perhaps a bit hard to translate into software. Because the mesh… Or perhaps not, are there ways of adapting to the situation without access to eye tracking sensors and other monitors of the physical world that the user is in?

P3:

Well, we do have lots of sensors in a phone. So, we have lots of sensors, like the gyroscope, the magnetometer, the accelerometer, several different cameras, including several on the backside for depth, but also several on the front side for facial recognition, and we have microphones. So, there are a bunch of sensors that potentially could be used to adapt the user interface to the current situation.

Not necessarily to the current state of mind of the user, but sometimes we could perhaps infer it. Yeah, through the voice, perhaps. So, there is information, and then the system can adapt to the current situation, but it could also adapt to where the user is on a development scale or something like that, which is not time critical here and now in the situation, but more over time, depending on log data instead.

I:

But then you would have to have long-term memory in these chatbots as well.

P3:

Yeah, you would need to have a user model of some sort, and it could be a very simple one. This user is on stage two. Yeah, currently that is not included. That is a really simple user model.

I:

I do not know if you can answer this one, but even outside the specific realm of chatbots, what risks should be considered when designing technology aimed at behaviour change or personal development?

P3:

Well, what are the risks? One risk is that you develop undesirable behavior. Another risk is that you do not get any development behavior. And the third risk is that you get dropouts, that people do not use it because it does not feel meaningful.

I:

How do you design for meaningfulness? Like, generally, what causes people to drop out when using technology?

P3:

Too much friction when it comes to usability and stuff like that is one. The thing is that people, if they feel that they cannot be bothered with it, or the hassle is too large, the cost is too large when it comes to hassle.

Another thing is that you need to perhaps develop a habit, and then you need to stick with it for a certain time to stop building that habit. And that is difficult. So, you need to keep the user in that way, like you do when you are trying to learn a language and use Duolingo. So, you have the notifications and reminders, and if you have turned off the notifications on your phone because you do not want to be distracted, then you sometimes get an email saying that we miss you or whatever.

So there are those things like notifications or reminders that you can use. Harri Oinos Kokkonen has a list of guidelines for behaviour change technologies, 19 guidelines or something like that. So that is probably something to look at. Well, that is the three risks that I see. So, developing the wrong behaviours, dropping out, or getting no effect. Dropping out and getting no effect is sort of related, but getting no effect is probably worse than dropping out. Because then you have wasted people's time. But wrong effect is probably even worse. That is the worst one. Getting no effect is still bad because you took up people's time without it getting any effect at all.

I:

So, one large potential risk then is if these technologies are underdeveloped but still put out to market and used by people for these purposes, then that could lead to the wrong behaviour change?

P3:

Yes, potentially. If you for instance use it for procrastination, that would be the wrong effect.